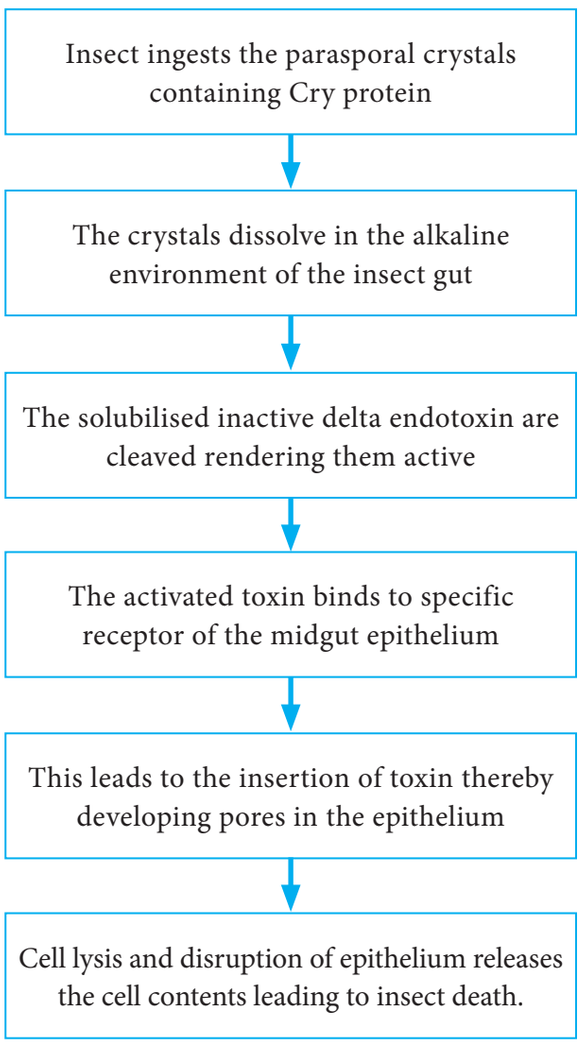
toxic to insects belonging to Lepidoptera, Coleoptera and other few insect orders.

Mode of action of Bt

The Bt cells sprayed on the leaves have to be ingested by the larval forms of the insects in order to exert its action. This is because the Bt toxin gets activated in the insect gut at a specific pH.

Process



Symptoms

- Larvae stops feeding
- Larvae becomes sluggish and static
- Water oozes out from the body
- Larvae dies and falls off the leaf

Various species of Bt are able to work against cotton boll worm, cabbage worm and gypsy moths.

Infobits

Photograph of a cotton plant showing opened and unopened bolls. BT cotton is a genetically modified cotton plant (GM crop) which has the gene for the crystal toxin integrated in its genome. Crystal toxin is expressed in plant parts which reduce the need for spraying pesticides. BT cotton is the only GM crop approved for commercial cultivation in India.

11.3.2 Fungal Biopesticides

These entomopathogenic fungi attack insects and cause diseases in insect body which lead to insect death. Two prominent fungi used as mycopesticide are

- *Beauveria bassiana* which causes white muscardine disease
- *Metarhizium anisopliae* which causes green muscardine disease

Mode of action of *Beauveria bassiana*

Beauveria bassiana, a filamentous fungus belongs to class Deuteromycetes also called imperfect fungi. It can be successfully used against Colarado potato beetle, (Figure 11.11) Codling moth and American boll worm.

This fungus invades the haemocoel of insects through spores. Once the spores attach to the cuticle, it germinates and the hyphae penetrates the insect cuticle (cuticle is the outer membrane of insects) Penetration is aided by formation of appresorium and penetration peg. The fungi

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